Integrating Textual Semantics with Visual Positioning: A Multi-Modal Approach to Enhanced Complex Table Understanding

Yujia Rong
The 32nd Research Institute of China Electronics Technology Group Corporation
Shanghai, China
yujiaarong@gmail.com

Bo Jiang
The 32nd Research Institute of China Electronics Technology Group Corporation
Shanghai, China
b26jiang@126.com

Hong Xu*
The 32nd Research Institute of China Electronics Technology Group Corporation
Shanghai, China
*Corresponding author:
znzx_xuhong@126.com

2025 6th International Conference on Computer Vision, Image and Deep Learning (CVIDL) | 979-8-3315-2324-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/CVIDL65390.2025.11085925

***Abstract*—In the field of project fund supervision, an important research topic is the intelligent data collection and analysis of contract information and financial statements. However, due to the high domain specificity and concise terminology usage in such documents, combined with the frequent occurrence of complexly styled tables, these factors present significant challenges for model-based data processing. This study proposes a multimodal feature fusion approach that integrates textual semantic features with visual positional features from document tables. Our model undergoes specialized pre-training on text-image alignment tasks, enabling simultaneous comprehension of tabular content while establishing explicit associations between textual elements and their spatial positions. The proposed visual-textual semantic feature fusion method has been validated on the publicly available FUNSD dataset. Experimental results demonstrate the effectiveness of our approach in addressing complex table document understanding tasks.**

Keywords-Visual-Textual Semantic Features Fusion; Complex Table Analysis; Table Structure Recognition; Data Extraction.

I. Introduction

With the rapid development of artificial intelligence technologies, data intelligence has become a core foundation for advancing industrial applications. In engineering practice, extracting analytically valuable data from large volumes of structured, semi-structured, and unstructured data, and constructing a data foundation based on these data, is a critical prerequisite for enabling practical value in industrial AI applications. In the context of project fund supervision, regulatory personnel frequently process document data such as contracts, procurement orders, bank transaction records, and financial statements. Due to commercial confidentiality requirements and legal compliance constraints, these documents are predominantly stored in paper form, necessitating manual data entry during internal regulatory processes. This workflow presents two significant issues: first, the low efficiency and high labor intensity of manual entry; second, the risks of privacy leakage, data fabrication, and challenges in data traceability associated with manual operations.

Furthermore, unlike the well-defined structured data in relational databases, the aforementioned documents generally exhibit highly unstructured characteristics. Notably, tables are widely used in these documents due to their high information density and intuitive expressiveness. Consequently, the identification and parsing of tables—particularly those with complex hierarchical or nested structures—have become a core challenge in the fields of document recognition and information extraction.

Based on the above context, the main contributions of this study can be summarized as follows:

1) This paper systematically reviews mainstream technical methods in the field of document recognition, covering three directions: scene text detection, complex table structure parsing frameworks, and dataset construction, while clarifying current bottlenecks in multimodal integration and structural perception.

2) This paper proposes a complex table parsing model based on bimodal feature fusion, innovatively combining textual semantic features and visual positional features to enhance the understanding of nested table structures.

3) The effectiveness of the proposed model is validated on the FUNSD public dataset, with comparisons conducted against the well-known open-source model SDMGR on the same dataset. Experimental results demonstrate the validity of our method.

II. Related Work

A. Scene Text Detection and Comprehension

As previously discussed, the original format of these documents is predominantly physical, with their digitized versions typically existing as scanned images or photographic records. Consequently, the input data processed by models are often in PDF or JPG formats. To streamline the workflow, researchers commonly utilize existing PDF conversion tools to segment documents into individual page images for model input. Therefore, the initial step in document parsing and complex table structure understanding can be fundamentally reduced to the problem of scene text detection and comprehension from images.

Addressing this challenge, Singh et al. identified a critical limitation in current optical character recognition (OCR) systems: the lack of real-world scene text annotations for training text detection and recognition models. To overcome this, they constructed the TextOCR dataset, which features annotated text of arbitrary shapes in real-world scenes. Based on this dataset, they trained an OCR model and further developed an end-to-end scene image text reasoning system named PixelM4C [1]. Experimental results demonstrated that the incorporation of

This research was funded by the National Key R&D Program of China (2024YFB 4504500).

Authorized licensed use limited to: South-Central Minzu University. Downloaded on April 22,2026 at 08:14:29 UTC from IEEE Xplore. Restrictions apply.

the TextOCR dataset significantly enhanced the cross-dataset performance of multiple OCR models, including PixelM4C.

In the domain of document semantics modeling, Weng et al. focused on relation extraction between words and clauses within texts. By constructing hierarchical graphs for words and clauses, their approach effectively reduced redundant information and improved the accuracy of relation extraction [2]. A related study by Cao et al. adopted a heterogeneous graph network to model intra-group data associations. Experimental evaluations on the NYT and WebNLG datasets yielded F1 scores of 93.2 and 94.7, respectively [3].

Furthermore, named entity recognition (NER) and relation extraction (RE) remain active research topics in this field. Hiraoka et al. introduced a table-based representation method for parsing unstructured text, leveraging context-aware word embeddings to model long-range dependencies between entities [4]. Wang et al., in contrast, drew inspiration from 2D table modeling techniques for entity interaction representation. Their architecture employs dual encoders: one for sequential NER representation and another for relational RE representation, interconnected through a collaborative mechanism to optimize joint performance [5].

B. Complex Table Structure Recognition Framework

For frameworks addressing table recognition and complex table structure parsing, numerous studies provide valuable insights. Microsoft's research team proposed TUTA (Table Understanding with Unified Pre-training Architecture), a unified pre-training architecture designed to model spatial layouts and hierarchical relationships in general structured tables [6]. The framework employs a two-dimensional coordinate tree structure to capture geometric constraints among table cells, offering a structural prior for subsequent semantic parsing.

In the domain of end-to-end document parsing, Li et al. developed the PPStructure V2 framework, which introduces a threshold-based YX sorting algorithm to resolve the text reading order modeling challenge in tabular data [7]. The team further advanced the vision-agnostic multimodal pre-trained model VI-LayoutXLM. Notably, although LayoutXLM was initially designed to integrate text, layout information, and image modalities, its visual backbone (ResNeXt-FPN) exhibited computational efficiency limitations during feature extraction. Experimental results revealed that removing the visual submodule did not significantly degrade performance on semantic entity recognition (SER) and relation extraction (RE) tasks; instead, performance improved on certain datasets. This suggests that VI-LayoutXLM serves as a more efficient and robust alternative in scenarios with low-quality images or ambiguous visual features [8].

Regarding multimodal document representation, Sun et al. proposed a dual-modal graph modeling approach. Their framework employs node encoders to extract features from textual and visual regions, while edge relationships model spatial adjacency between adjacent text units[9]. Chi et al. innovatively applied graph neural networks to PDF table structure recognition, developing the GraphTSR model. This framework leverages graph convolution mechanisms to capture topological relationships among table cells, with its effectiveness validated on the SciTSR dataset [10].

Traditional machine learning and deep learning approaches remain relevant. Prasad et al. designed CascadeTabNet, a single CNN-based architecture for cascaded table detection and recognition [11]. Qiao et al. introduced a soft pyramid mask learning framework that integrates local features with global contextual information, demonstrating enhanced robustness in table structure recognition tasks [12].

C. Datasets

To advance the development of table structure recognition technologies and enhance model performance, the construction of high-quality datasets has become another critical research direction. IBM's Nassar et al. developed SynthTabNet, a synthetic table structure dataset generated through generative models, based on existing datasets such as PubTabNet, FinTabNet, and TableBank [13]. This dataset not only contains 600,000 high-diversity table images but also provides high-precision synthetic annotations (ground truth), establishing a foundational basis for reproducible research. The FinTabNet subset within SynthTabNet is derived from annual financial reports of S&P 500 companies, encompassing extensive financial domain-specific terminology, which is vital for building domain-adaptive models.

Another IBM team developed and released the large-scale PubTabNet dataset, comprising over 568,000 heterogeneous table images extracted from scientific papers in PDF format [14]. Furthermore, Smock et al. constructed a more comprehensive table extraction dataset named PubTables-1M. This dataset includes nearly one million tables from scientific literature, supports multiple input modalities, and provides detailed table header information and positional annotations, making it applicable to various modeling approaches [15].

In consideration of limitations in existing annotation methodologies, Yang et al. highlighted that conventional bounding box (Bbox) or quadrilateral annotations struggle to adapt to the structural variations of complex tables [16]. To overcome this, they proposed TabRecSet, a large-scale table recognition dataset employing polygon annotations. The dataset comprises 38,100 table images (20,400 English tables and 17,700 Chinese tables), with data sources spanning diverse real-world scenarios, including scanned images, photographed images, document-based tables, financial invoices, Excel spreadsheets, and educational examination papers.

III. Core Techniques and Methods

A. Overall Framework

The overall framework of the proposed method is illustrated in Figure 1. The original table data is stored and input into the model in image format. For textual semantic feature extraction, we first employ a general OCR tool to recognize text information from the image. The unsorted and unsegmented text is then fed into the InfoXLM model [17]. InfoXLM is a cross-lingual pre-trained model that enhances its ability to transfer knowledge across languages through joint training on multiple pre-training tasks, including multilingual masked language modeling,

Authorized licensed use limited to: South-Central Minzu University. Downloaded on April 22,2026 at 08:14:29 UTC from IEEE Xplore. Restrictions apply.

translation language modeling, and cross-lingual contrastive learning. We believe that the textual feature vectors derived from processing text information using the pre-trained InfoXLM model can improve the robustness of our model in textual semantic understanding. For visual features, we utilize a CNN encoder to capture visual feature information from the table image [11]. Subsequently, we apply block-diagonal tensor decomposition via the Kronecker product to fuse the textual feature vectors and visual feature vectors, thereby modeling the higher-order correlations between textual and visual features. Finally, the feature matrix is input into the Transformer Encoder and passed through a BIO tagging classification layer to output the category labels for each part of the table.

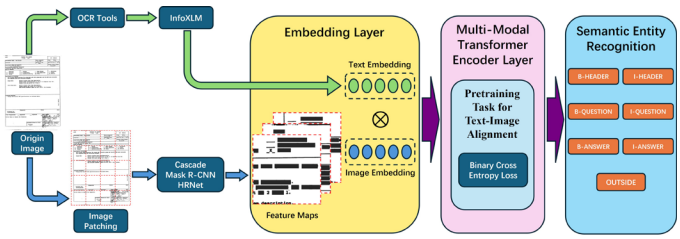

Figure 1. Overall framework of text-image features fusion.

B. Fusion of Textual and Visual Features

As described in the previous section, we employ a pre-trained InfoXLM model [17] to directly obtain textual feature vectors and use a CNN encoder [11] to extract feature maps from the image. Specifically, we first divide the original image (800×1000) into 200×200 patches. These patches are then processed using a Cascade Mask R-CNN HRNet-based encoder to generate the image's feature maps. In CNNs, feature maps are high-dimensional tensors output by convolutional layers. To convert them into lower-dimensional vector representations, we apply average pooling to reduce the feature maps to fixed-size outputs and subsequently flatten them into an embedding sequence, thereby obtaining the visual feature vector of the original image.

For the fusion of textual and visual features, we adopt the block-diagonal tensor decomposition approximation of the Kronecker product. The computation of the Kronecker product is defined as:

$$\mathrm{A} \otimes \mathrm{B} = \begin{bmatrix} a_{11}\mathrm{B} & a_{12}\mathrm{B} & \cdots & a_{1n}\mathrm{B} \\ a_{21}\mathrm{B} & a_{22}\mathrm{B} & \cdots & a_{2n}\mathrm{B} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}\mathrm{B} & a_{m2}\mathrm{B} & \cdots & a_{mn}\mathrm{B} \end{bmatrix} \quad (1)$$

This operation explicitly models all pairwise interactions between the two modalities, capturing higher-order correlations between textual and visual features. In multimodal learning, directly computing the Kronecker product of textual and visual features preserves complete interaction information but may lead to a significant increase in feature dimensionality. Due to the high-dimensional nature of the Kronecker product, direct application in model training and inference would sharply escalate computational complexity and memory consumption. To address this, we employ block-diagonal tensor decomposition to approximate the original computation and achieve dimensionality reduction. The objective of block-diagonal tensor decomposition is to decompose a high-order tensor $\mathcal{X}$ into the following form:

$$vec(\mathcal{X}) \approx \sum_{r=1}^{R} \lambda_r \cdot \left(A_N^{(r)} \otimes \ldots A_2^{(r)} \otimes A_1^{(r)}\right) \cdot vec(\mathcal{G}_r) \quad (2)$$

Here, λ_r denotes the scaling factor, $\mathcal{G}_r$ is the block-diagonal core tensor, and $A_N^{(r)}$ represents the factor matrix for the r-th component. By leveraging the block-diagonal structure, non-zero elements of $\mathcal{G}_r$ are restricted to diagonal blocks, thereby reducing the parameter scale during computation.

C. Cross-Modal Pre-training for Text-Image Alignment

Text-Image Alignment is a critical pre-training task in our model, designed to enable the model to learn spatial positional correspondences between text and images. We adopt the same pre-training strategy as described in reference [8], which involves randomly selecting certain text lines and masking their corresponding image regions. For each text token, the model predicts a binary label Covered or Not Covered based on whether its associated image region is masked. The model parameters are optimized using the mean binary cross-entropy loss $\mathcal{L}_{mean}$, defined as:

$$\mathcal{L}_{mean} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \, log(\hat{y}_i) + (1 - y_i) \, log(1 - \hat{y}_i)] \quad (3)$$

Through this Text-Image Alignment pre-training task, we optimize the parameters of the Transformer Encoder, enhancing the model's understanding of text-image correspondences, particularly under scenarios involving partial occlusion or degraded image quality.

IV. Experiments

A. Datasets

The text-image alignment pre-training task of our model was conducted on the IIT-CDIP dataset. Subsequently, we performed fine-tuning on the open-source FUNSD dataset [18]. The FUNSD dataset employs the BIO annotation scheme, where the forms originate from diverse domains such as marketing, advertising, and scientific reports, enhancing the dataset's diversity and generalization capability. All forms in the dataset are single-page, low-resolution, rasterized documents that are subject to real-world noise, reflecting the actual conditions of documents in practical scenarios.

B. Experimental Results and Analysis

In the annotation framework of the training dataset, target text is categorized into four core classes: Header (table title), Question (table subheading), Answer (corresponding attribute value), and Other (unclassified content). Specifically, the Header class refers to the table title, the Question class corresponds to the table subheading, the Answer class represents the specific attribute values associated with the Question class, and the Other class encompasses remaining content not explicitly classified. This study aims to train a model capable of accurately identifying key components within complex table structures, thereby providing foundational support for subsequent intelligent information extraction tasks. As shown in Figure 2, the experimental results demonstrate that the model successfully achieves the intended objectives.

Authorized licensed use limited to: South-Central Minzu University. Downloaded on April 22,2026 at 08:14:29 UTC from IEEE Xplore. Restrictions apply.

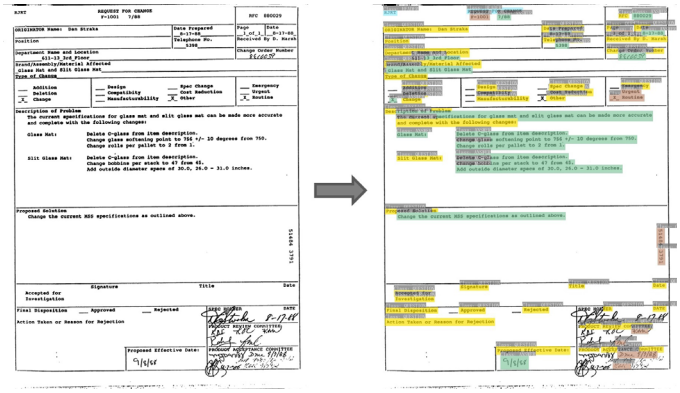

Figure 2. Recognition result of our model.

The model's recognition results are visualized using color-coded annotations: blue regions indicate identified table headers (Header), yellow regions represent detected subheadings (Question), green regions correspond to recognized attribute values (Answer), red regions denote other detected content (Other), and gray annotations reflect the class labels assigned by the model to each segment.

Table 1 Evaluation Metrics of Experimental Results

Models	EPOCHs	Precision	Recall	Hmean
our	200	84.61%	85.34%	84.98%
SDMGR	200	70.98%	72.12%	71.54%

Since table structure parsing requires balanced consideration of precision and recall, we adopt the F1 Score as the core evaluation metric for model performance. As shown in Table 1, after training for 200 epochs on the FUNSD dataset, the proposed model achieves an F1 Score of 84.98% on the validation set. To validate the model's effectiveness, we selected the open-source SDMGR model as a baseline. While SDMGR demonstrates strong performance on the WildReceipt dataset (primarily consisting of receipts) with an F1 Score of 88.7%, it only achieves 71.54% on the more complex FUNSD dataset. Comparative experiments confirm that our model exhibits significant advantages in handling intricate table structure comprehension tasks.

V. Conclusions

This paper systematically reviews the advancements in mainstream technical methods for complex table structure parsing in document recognition, covering three key areas: scene text detection, complex table parsing frameworks, and dataset construction. It also identifies the primary challenges in current research. In response to the limitations of existing methods in handling nested table structures, this study proposes a bimodal feature fusion-based parsing model that integrates textual semantic features with visual features, significantly enhancing the model capability for complex table structures. Experimental results demonstrate that the proposed model outperforms the mainstream open-source model SDMGR on the FUNSD public dataset, validating its effectiveness.

Future work will focus on explicitly modeling the intrinsic table structure itself, incorporating structural features as a new dimension into the model. Additionally, further exploration will be conducted to improve the model's ability to understand textual semantics, thereby enhancing the generalization capability of the proposed approach.

Acknowledgment

This work was supported by the National Key R&D Program of China (Grant No. 2024YFB4504500).

References

[1] Singh, A., Guan Pang, Toh, M., Jing Huang. (2021). TextOCR: Towards large-scale end-to-end reasoning for arbitrary-shaped scene text. In: CVPR 2021. pp. 8798–8808.

[2] Weng Y., Xu B., Cai R. (2024) Fine-grained multi-document summarization extraction based on heterogeneous graph hierarchical learning. J. COMPUTER ENGINEERING, 50(3): 336-344.

[3] Cao Y., Cheng Y., He Z., Xu K., Yan J., Li Y. (2024) Document Context Heterogeneous Representation for Sentence-Level Relation Extraction. J. COMPUTER ENGINEERING, 50(5): 111-119.

[4] Ma, Y., Hiraoka, T., Okazaki, N. (2020). Named Entity Recognition and Relation Extraction using Enhanced Table Filling by Contextualized Representations. arXiv.

[5] Jue Wang, Wei Lu. (2020). Two are Better than One: Joint Entity and Relation Extraction with Table-Sequence Encoders. arXiv.

[6] Wang, Z., Dong, H., Jia, R., Li, J., Fu, Z.. (2021). TUTA: Tree-based Transformers for Generally Structured Table Pre-training.

[7] Li, C., Guo, R., Zhou, J., An, M., Du, Y., Zhu, L., Liu, Y., Hu, X., Yu, D. (2022). PP-StructureV2: A Stronger Document Analysis System.

[8] Xu, Y., Xu, Y., Lv, T., Cui, L., Wei. (2021). LayoutLMv2: Multi-modal pre-training for visually-rich document understanding. In: ACL-IJCNLP 2021. Proceedings of the Conference. pp. 2579–2591.

[9] Sun, H., Kuang, Z., Yue, X., Lin, C., & Zhang, W. (2021). Spatial dual-modality graph reasoning for key information extraction. ArXiv.

[10] Zewen Chi, Heyan Huang, Heng-Da Xu, Houjin Yu, Wanxuan Yin, Xian-Ling Mao. (2019). Complicated table structure recognition. arXiv.

[11] Prasad, D., Gadpal, A., Kapadni, K., Visave, M., Sultanpure, K. (2020). CascadeTabNet: an approach for end to end table detection and structure recognition from image-based documents. arXiv.

[12] Qiao, L., Li, Z., Cheng, Z., Zhang, P., Pu, S., Wu, F. (2021). LGPMA: Complicated Table Structure Recognition with Local and Global Pyramid Mask Alignment. In: ICDAR 2021. pp. 99–114.

[13] Nassar, A., Livathinos, N., Lysak. (2022). TableFormer: table structure understanding with transformers. In: CVPR 2022. pp. 4604–4613.

[14] Xu Zhong, ShafieiBavani, E., Jimeno Yepes, A. (2020). Image-Based Table Recognition: Data, Model, and Evaluation. In: ECCV 2020.

[15] Smock, B., Pesala, R., Abraham, R. (2022). PubTables-1M: Towards comprehensive table extraction from unstructured documents. In: CVPR 2022. pp. 4624–4632.

[16] Yang, F., Hu, L., Liu, X., Huang, S., Gu, Z. (2023). A large-scale dataset for end-to-end table recognition in the wild. arXiv.

[17] Chi, Z., Dong, L., Wei, F., Yang, N., Singhal, S. (2021). INFOXLM: An Information-Theoretic Framework for Cross-Lingual Language Model Pre-Training. In: NAACL-HLT 2021. pp. 3576–3588

[18] Jaume, G., Kemal Ekenel, H., & Thiran, J.-P. (2019). FUNSD: a dataset for form understanding in noisy scanned documents. In: ICDARW 2019

Authorized licensed use limited to: South-Central Minzu University. Downloaded on April 22,2026 at 08:14:29 UTC from IEEE Xplore. Restrictions apply.